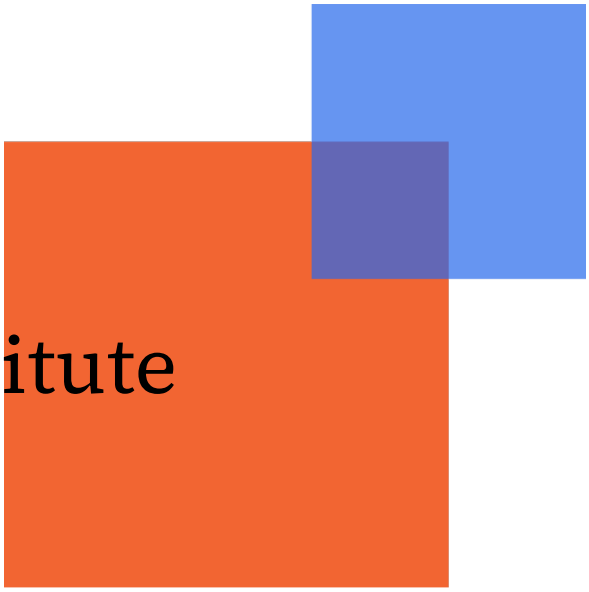
Runaway Citations and the Persistence of Bitcoin Misinformation

Working Paper

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23/08/2024

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Executive Summary

Bitcoin misinformation has firmly entrenched itself in the discourse about Bitcoin and energy. Misinformation generally results from a low publication standard – methodological poverty is consistent in work with a negative sentiment. The archetypes of these publications continue to build a significant count of citations despite using flawed methods. However, methodologically defensible work has begun to shift the narrative by demonstrating accurate, albeit highly specific, results.

Where Does Misinformation in Bitcoin Come From?

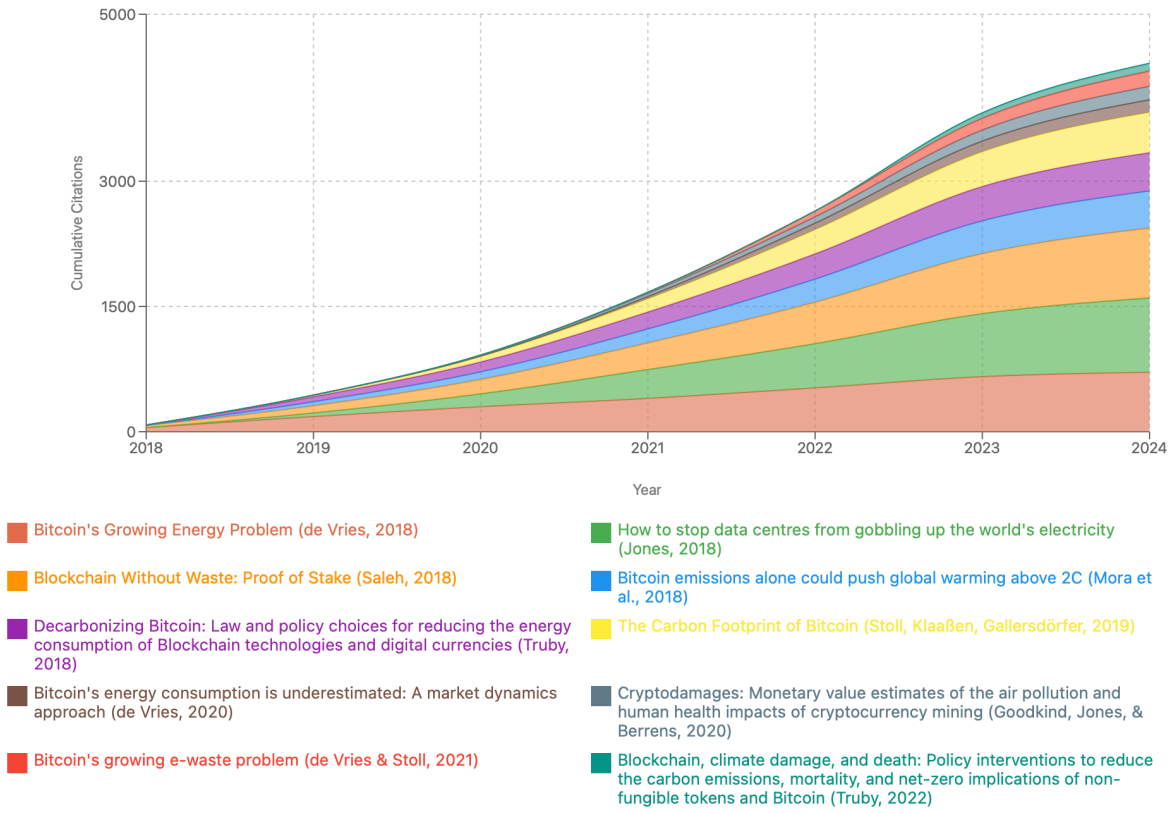
Misinformation has become a persistent element in discussions about Bitcoin's energy consumption. Flawed research, lacking methodological rigor, has been the breeding ground for misinformation that continuously circulates within the discourse, creating a challenge for those aiming to present a more balanced view. These findings, despite their flaws, have significantly shaped the broader discourse on Bitcoin, creating a narrative that disproportionately highlights the negative aspects without considering the benefits or addressing the misinformation. Despite a recent emergence of methodologically sound research, the established narrative remains deeply rooted.

The consequences of false narratives are not just theoretical - they have led to real-world actions such as regulatory measures (Riley, 2021), mining restrictions (Xie, 2019), and the stigmatisation of Bitcoin (Kinney, 2021a). This persistence hampers the potential of Bitcoin to contribute positively to financial inclusion, renewable energy deployment, electrical grid management, and methane mitigation. We can measure the spread of widely debunked articles and their influence on the field by examining the citation counts.

Key contributors to this problem include Alex de Vries (2018, 2019, 2020, 2020, 2021, 2023), whose works have amassed over 1,130 citations, and his collaborators Christian Stoll (2022; 2021; 2020, 2021; 2019, 2023) and Lena Klaaßen, (2022; 2020; 2019, 2023) with nearly 1,000 and 800 citations, respectively. De Vries's most influential work, "Bitcoin's Growing Energy Problem" continues to be cited in new

publications despite being thoroughly debunked by peer-reviewed studies (Sai & Vranken, 2023) – and it's hardly the only paper of its kind. The chart below tracks the cumulative citations of just ten influential, but inaccurate articles over time – each of which cites the original misinformation paper – de Vries (2018). The graph illustrates how these works, themselves utilising flawed methodologies and therefore making unsubstantiated claims, have gained traction and continue to influence the field. Many of these papers don't conduct original studies but rather compile and propagate existing misinformation, creating a self-reinforcing cycle of inaccuracy. New researchers entering the field are immediately exposed to these flawed studies, making it increasingly difficult for accurate information to take hold. Misinformation is a very resilient thing, and once established in discourse, it's tough to remove it entirely.

Cumulative Citations for Publications with Methodological Flaws 2018 – 2024





The following table summarises the flaws in the papers tracked in the chart above.

Article	Issues
de Vries, (2018)	• Doesn't build on existing theory and provides its methods without positioning it within any other field. • Oversimplifies mining and underestimates the lifespan of hardware. • Uses anecdotes to back up claims.
Jones, (2018)	• Bases claims on flawed de Vries 2018. • Is not a study but rather an accumulation of convenient anecdotes.
Saleh, (2018)	• Bases its energy use assumptions on de Vries 2018 and, as such, are overstated.
Mora et al., (2018)	• Assume participants would use unprofitable hardware. • Holds efficiency assumptions static for 100 years of projection. provides no apparent reasoning behind this choice, • Random assignment of mining hardware, • Uses IP of mining pool as the location of miner, • Uses outdated carbon intensity data.
Truby, (2018a)	• Bases energy consumption data on de Vries 2018. • Calculates energy use on a per-transaction basis.
Stoll, et al., (2019)	• Makes a range of assumptions to generate their findings including using IPO filings to estimate mining hardware distribution (Public miners are a minor proportion of all companies and relatively unique in approach). • Assumes sales volumes of hardware remain consistent. • Uses the location of a single pool to represent all pools for Europe and China. • A single interview assumes power use effectiveness for all miners.
de Vries, (2020)	• Assumes old hardware (S9) is still dominant in 2019. • Uses arbitrary distribution of sales between manufacturers without evidence. • Does not justify power usage effectiveness.
Goodkind, et al., (2020)	• Assumes emissions for whole on-grid electricity generation are equivalent to the mix used by bitcoin mining. • It uses De Vries and Digiconomist (which is also De Vries) to estimate Bitcoin's power use.
De Vries & Stoll, (2021)	• Calculates mining hardware's lifespan as 1.29 years by assuming demand peaks within months of release and then rapidly declines. • Later, a chart estimates four years of sales for the S9. • Assumes a static price for electricity of USD 0.05. • Applies flawed logic to a per-transaction measure of e-waste. • Assumes there's no value in recycling hardware.
Truby, (2022a)	• Uses de Vries to base energy and emissions rate. • Estimates of deaths per year are based on flawed emissions figures, but no methods are provided for that calculation.

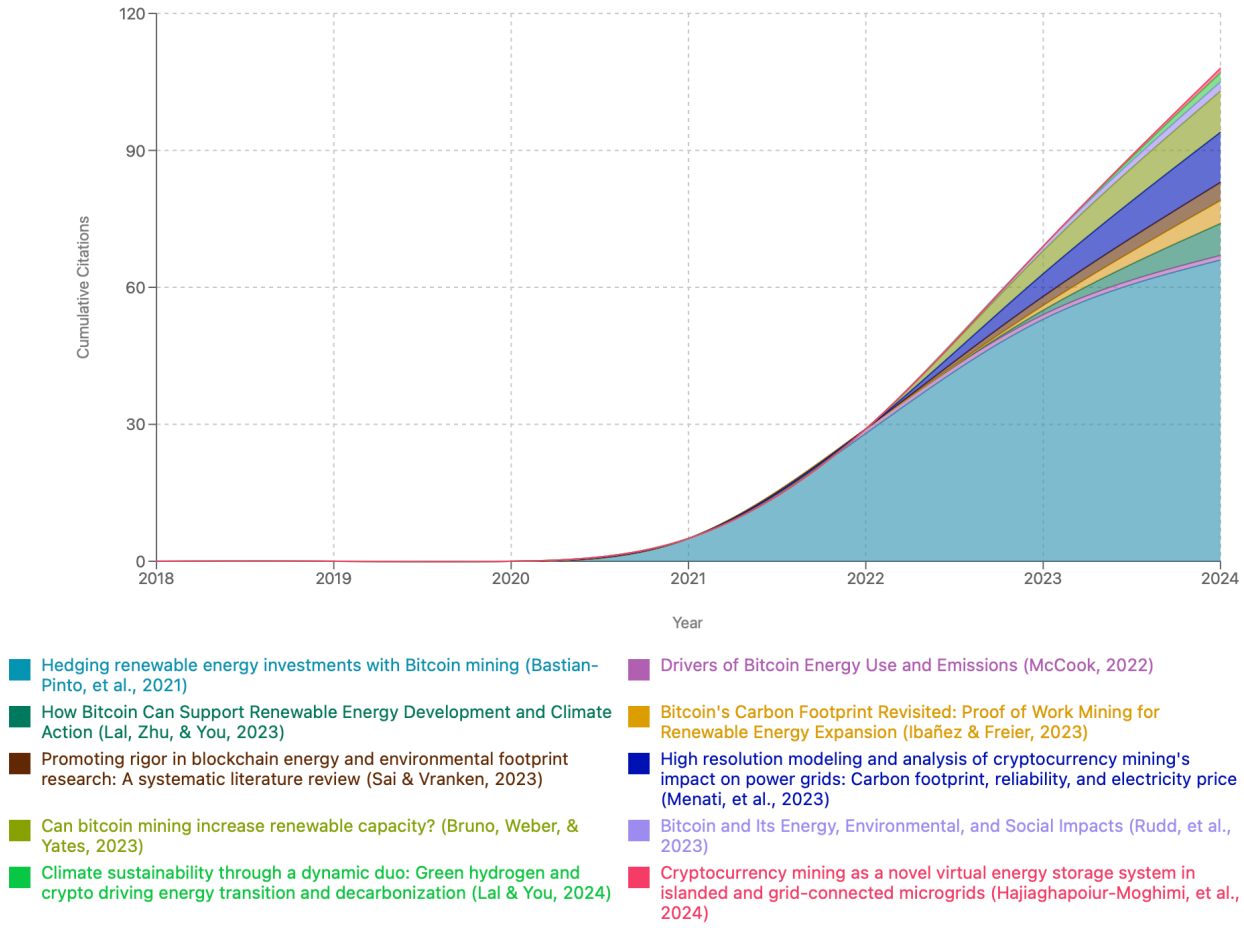
What Does It Mean For Bitcoin?

The implications of this chart and the table are that nearly 5,000 faulty papers have been written about Bitcoin. It's important to note that this is only in academia and from a single starting point. Mainstream media picks up scientific literature and reports it often in simplified terms. According to Sai and Vranken's (2023) systematic review, this is at a rate of approximately 10:1. So there is something on the order of 450 articles that extend an unsubstantiated narrative to the significantly broader mainstream audience. This reach extends the influence of these articles without the qualifying logic and essential context that comes from articles subjected to peer review before publication.

What Can Be Done About Misinformation?

To avoid having poor quality publications continue to influence the knowledge base, it's essential to promote research that adheres to rigorous standards, ensuring that data is verifiable, and results are replicable. Although these studies are less likely to produce sensational headlines, their importance lies in their accuracy and reliability. Over time, as these truthful narratives gain more exposure, they can begin to dilute the misinformation that currently dominates the conversation. We can find truth in articles that follow accepted methods, provide access to their sources of data and allow for replication, or the cross-checking of their results. And this truth is beginning to emerge. The chart below tracks citations for research that methodologically examines Bitcoin energy use. These studies either use verifiable and reproducible data or perform a systematic review of literature, all with published and verifiable methods.

Cumulative Citations for Studies with Appropriate Methodologies 2018 - 2024



As you can see, these studies' citation count is a mere one-forty fifth of the misinformation group. This small count is the impact of three years of head start in controlling the narrative. It's also important to note that because a study examines a particular phenomenon in a specific context, it can't make the same broad, sweeping claims that others might. Claims such as Bitcoin emissions could be responsible for increasing the earth's temperature by 2°C or that Bitcoin is directly responsible for consuming gigalitres of water because a coal powerplant causes it to evaporate. Or that mining hardware is only usable for 1.29 years before being disposed of. Or that in 2018 Bitcoin mining was responsible for more than 350 deaths in the USA and China. The standard for claims is much higher in a peer-reviewed and published empirical

study; therefore, these claims are less grandiose and more couched in the situation under study. For example, consider the key findings from two studies:

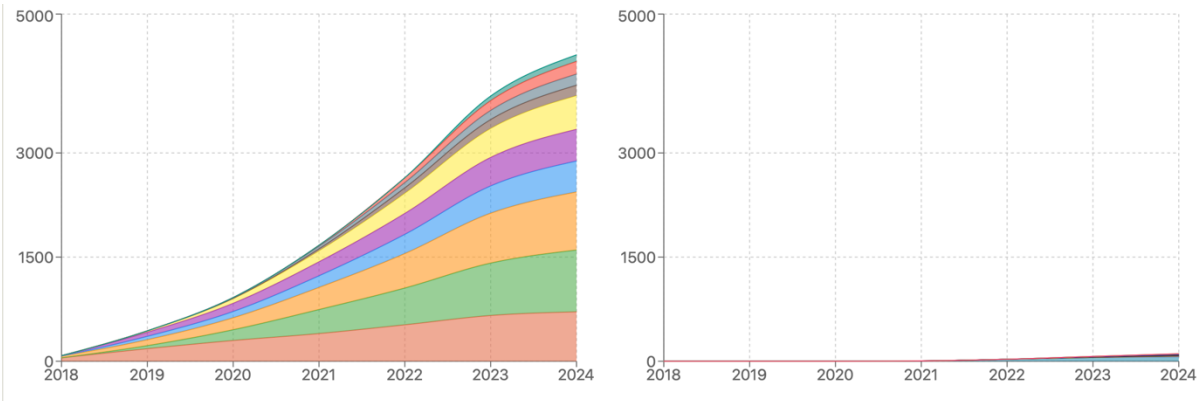
> *"[Using bitcoin mining] to store excess renewable energy as the developed cryptocurrency for consumption in another period underscores its potential as an efficient and cost-effective solution for energy management within the [microgrid]"* (Hajiaghapour-Moghimi et al., 2024)

Or:

> *"Therefore, modelling [bitcoin mining] flexible demand is shown not to be detrimental to power grid reliability even with significant amounts at certain locations."* (Menati et al., 2023)

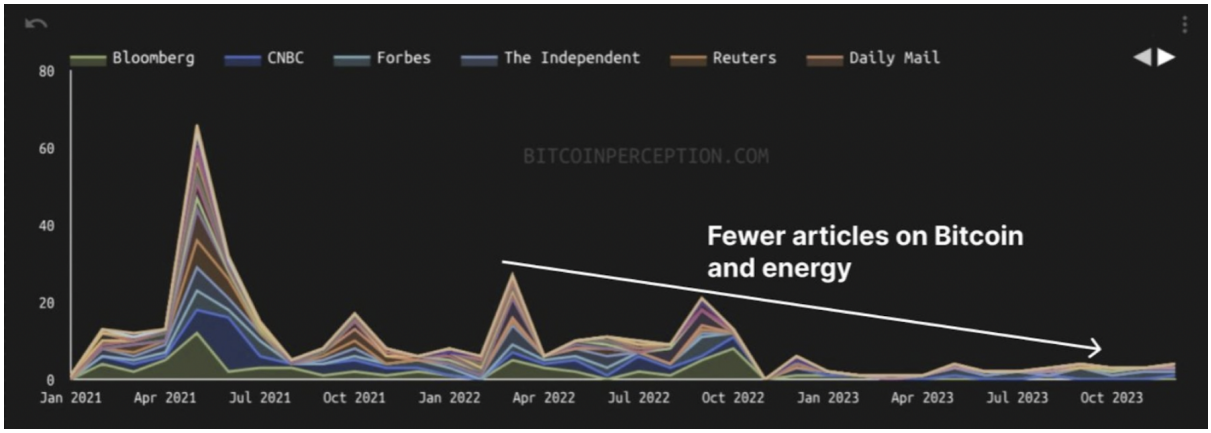
The soundbites don't exactly write themselves, so the barriers are twofold. First, the results of high-quality studies don't lend themselves to exciting and eye-catching headlines. Second, the emergence of rigorous studies has only really begun in the last couple of years. Side by side, the discrepancy is even more glaring and demonstrates the impact of a three-year head start on controlling the narrative, whether or not your methods are valid.

Side-by-Side Comparison of Cumulative Citations



Despite the lack of a head-start and the requirement for circumspection and accuracy, we can already see the effect of empirical work on the narrative around Bitcoin and energy. Since 2022 (the time that accurate studies began emerging), the publication of media on Bitcoin and energy has fallen significantly. This trend is an excellent example of how robust even the relatively specific findings of high-quality science can be in combatting the tide of entrenched misinformation.

Mainstream Articles on Bitcoin and Energy



What is significant about the distinction between the two sets of publications? If the two bodies of literature agreed, that would be one thing, but they most certainly do not. The papers stemming from de Vries are overwhelmingly negative in their sentiment. Conversely, publications conducting real-world studies have findings showing that the effects of Bitcoin and mining are almost exclusively positive. We'll let the reader decide what the implications of that may be.



Conclusion

Introducing resilient truths into the discourse is crucial. This process involves building an unbiased foundation of knowledge through rigorous, independent research. By carefully digesting this research, publishing clear and accessible results, and fostering open discussions, we can slowly but surely change the composition of the discourse. This process takes time and effort. It requires patience, persistence, and a commitment to academic integrity. But with each factual study published, each misconception addressed, and each nuanced discussion had, we're adding new, persistent knowledge to the Bitcoin discourse ecosystem.

The goal should be to avoid overcorrecting with floods of pro-Bitcoin sentiment. Instead, we need to restore balance—we need to create a healthy ecosystem where facts can thrive, myths are challenged, and genuine concerns are addressed thoughtfully. In this balanced environment, Bitcoin can then be evaluated accurately for its benefits and flaws. Through this work, we're cultivating a thriving, diverse ecosystem of knowledge that can adapt, evolve, and flourish.

Post-Script:

The author rebukes several authors and their articles in this publication, which will be referenced in the following section. I am aware that this is an irony. However, in the interest of demonstrating this significance and importance, I have deemed the extra citation for each article a necessary means to an end. It is hoped that in the future, authors will consider the impact of including these and other articles that may constitute misinformation.



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